Rahul Muralidhara Koushik

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Education

- **Arizona State University** — Tempe, USA
  Master's of Science - Computer Science — *Aug 2021 - May 2023*
- **PES University (PESU)** — Bengaluru, India
  Bachelor of Technology - Computer Science and Engineering — *Aug 2017 - Jul 2021*

Skills

- **Languages**: Python, Go, C++, JavaScript, SQL, Swift
- **AI & Agent Systems**: Multi-agent workflows, Retrieval-Augmented Generation (RAG), Vector DB, LLMs, AI Automation
- **Backend, Cloud & Infrastructure**: Git, REST APIs, gRPC, Kubernetes, Docker, Kafka, AWS, Linux, CI/CD (Jenkins)
- **UI**: React, Next.js, Tailwind CSS, Swift (iOS), Firebase

Experience

- **Hewlett Packard Enterprise** — San Jose, US
  Cloud Software Engineer — *Jun 2023 - Present*
  - **Observability, Monitoring and Analytics:**
    - Spearheaded an observability platform at scale (**30K devices**, **10K customers**) processing millions of logs
    - Enabled log-level RCA across **hardware, OS, backend, UI, and cloud**, reducing resolution time from **30 min** to **3 min**
    - Visualized failure trends, improving first-time success from **55% to 85%** and enabling data-driven fix prioritization
  - **Serviceability: Setup and Data migration**
    - Accelerated the setup process for new users by **8%** last quarter by developing observability initiatives for storage devices, including intuitive setup blueprints; collaborated with cross functional PCBE and DHCI teams
    - Orchestrated and integrated **6 full-stack Peer Motion features** enabling data migration and autonomic replication across heterogeneous storage systems (**3PAR, Nimble, Alletra MP, Alletra 9K**) with FC and iSCSI domain support
    - Added **domain, multi-link and VLAN tagging** support for Data migration flows
  - **INNOVATION**:
    - Built a **Documentation Automation Agent** using an agent-backed vector knowledge base to auto-update docs per PR, generate API, update Confluence docs saving **5 engineer-hours/week per team**
    - Proposed a **cross-Business Unit** observability and traceability framework integrating telemetry from **Storage** (Alletra), **Networking** (Aruba / Juniper), and **Server** platforms—laying the foundation for **360°** ecosystem visibility, unified traceability, and enhanced customer experience insights *(2025 initiative)*
    - **RAG-Based AI PDF bot** [2024 Hackathon Winner] - Engineered an end-to-end Retrieval-Augmented Generation chatbot ("Ask PDFs Anything"), enabling intelligent querying and contextual insight extraction from PDFs.
- **The Luminosity Lab** — Tempe, US
  Research Aide — *Feb 2022 - May 2023*
  - Led an iOS development team with a goal to revamp the iOS and Android platform Apps while helping the clients make mobile product decisions - **https://thetouchpointsolution.com/**
  - Employed HealthKit functionality to store HRV & other sensor values to classify users into six primary stress personalities
  - Designed a telehealth solution that allows doctors to start remote meetings with patients which saves **15 minutes** per visit
- **Aruba : Hewlett Packard Enterprise** — Tempe, US
  Software Engineering Intern — *May 2022 - Aug 2022*
  - Created a test automation tool to configure network topologies saving an estimated **150k hours** of engineering time/year
  - Conducted product presentation with panels comprising senior engineers and cross-functional representatives
- **Webex by Cisco** — Bengaluru, India
  Technical Undergraduate Intern — *Jan 2021 - Jul 2021*
  - Implemented a variety of APIs for scheduling calls and meetings for Cisco Webex **iOS SDK v3**
  - Developed a sample iOS App called **Kitchen Sink** that demonstrates the usage the iOS SDK APIs

Key Projects

- **AI-Driven YouTube Automation Agent (Multi-Agent System)**: Engineered a multi-agent AI system for fully automated YouTube content creation and publishing, coordinating 6+ specialized agents for content planning, audio generation, visual creation, video composition, metadata optimization, and uploads; triggered via a single Python command and capable of producing 100+ videos per execution
- **Fetch Rewards**: This is a sample scalable swift Application that displays 25+ categories of food, types and 2000+ recipes in a user-friendly manner. The network manager handles all kinds of APIs from https://www.themealdb.com/api.php
  Demo Video sample - **Video**